

Lab 02: IP Subnetting and Variable Length Subnet Mask (VLSM)

Duration: 1 hour 30 minutes | In-Lab Assessment

Tools: Cisco Packet Tracer

Objectives

By the end of this lab, students will be able to:

- Understand the basics of IPv4 subnetting and CIDR notation.
- Subnet a fixed-length network to meet given host requirements.
- Apply Variable Length Subnet Mask (VLSM) to allocate address blocks efficiently.
- Design and implement a multi-LAN topology in Cisco Packet Tracer using VLSM.
- Configure classless routing using RIP version 2 and explain the purpose of disabling automatic summarization.

Lab Description

Part A: Subnet (Subnetwork)

A subnet is a logical, visible subdivision of an IP network. The practice of dividing a network into two or more smaller networks is called subnetting. This improves network efficiency, security, and manageability by:

1. Reducing Broadcast Traffic: Keeping local traffic within the subnet and confining broadcast domains.
2. Improving Security: Isolating different parts of the network from each other.
3. Efficient Address Allocation: Preventing large-scale waste of IP addresses.

Part B: IPv4 Addressing and Subnet Masks

An IPv4 address is a 32-bit number, usually represented in dotted-decimal notation (e.g., 192.168.1.1). It is divided into two parts: the Network ID (identifying the entire network/subnet) and the Host ID (identifying a specific device within that network).

The Subnet Mask is a 32-bit mask used to determine which part of the IP address is the Network ID and which part is the Host ID. The bits in the mask that are '1' correspond to the Network ID, and the bits that are '0' correspond to the Host ID.

Part C: CIDR (Classless Inter-Domain Routing) Notation

CIDR notation, also known as slash notation (e.g., /24), is a compact way to represent the subnet mask. The number after the slash indicates the number of bits in the IP address that are dedicated to the Network ID/Subnet ID. For example, a mask of /24 means the first 24 bits are for the network, leaving $32 - 24 = 8$ bits for the host addresses.

Part D: Variable Length Subnet Mask (VLSM)

VLSM is a subnetting technique that allows the use of different-sized subnets within the same network. Instead of creating a set of equal-sized subnets (Fixed-Length Subnet Mask or FLSM), VLSM allows you to size each subnet exactly according to its unique host requirements.

Benefit: VLSM dramatically increases IP address utilization efficiency, especially for point-to-point links which only require two usable host addresses.

Part E: Basic IPv4 Subnetting

1. Theory Review:

- Recall that a subnet mask of $/n$ means the first n bits are network/subnet and the remaining $32 - n$ bits host.
- Compute # of subnets: 2^Δ , where Δ is the number of bits borrowed.
- Compute hosts per subnet: $2^{\text{host bits}} - 2$, where the first address is the network address and the last address is the broadcast address.

2. Exercise: Given network 172.16.0.0/16, create four equal-sized subnets.

- Determine how many bits to borrow for 4 subnets.
- Write each subnet's network address, subnet mask (in dotted decimal and CIDR), usable host range, and broadcast address.

Part F: Variable Length Subnet Mask (VLSM) Design

1. Host Requirements Table:

Subnet	Required Hosts
netA	14
netB	28
netC	2
netD	7
netE	28

2. Design Steps:

- Order subnets by descending host requirement.
- For each, choose the smallest CIDR block that accommodates the required hosts.
- Allocate subnets sequentially from 204.15.5.0/24, incrementing network address accordingly.

3. Resulting VLSM Plan:

Subnet	Network	Mask	Usable Hosts Range
netB	204.15.5.0	/27(255.255.255.224)	204.15.5.1 – 204.15.5.30
netE	204.15.5.32	/27	204.15.5.33 – 204.15.5.62
netA	204.15.5.64	/28(255.255.255.240)	204.15.5.65 – 204.15.5.78
netD	204.15.5.80	/28	204.15.5.81 – 204.15.5.94
netC	204.15.5.96	/30(255.255.255.252)	204.15.5.97 – 204.15.5.98

Note: The last address in each subnet is reserved as the broadcast address and is not usable for host assignment.

Part G: RIP Configuration and Classless Routing

To enable routing between the subnets defined above, configure RIP version 2 with the following guidelines:

1. On each router, enter configuration mode and enable RIP:

```
router rip
  version 2
  no auto-summary
  network 204.15.5.0
```

2. **Why RIP Version 2?**

RIP version 1 only supports classful routing and does not advertise subnet masks. Since this lab uses VLSM (variable-length subnet masks), it is essential to use RIP version 2, which supports classless routing and includes subnet mask information in its updates.

3. **Why no auto-summary?**

By default, RIP summarizes routes to their classful boundaries (e.g., summarizing all subnets under 204.15.5.0/24). This behavior can cause incorrect routing in discontinuous networks. Disabling automatic summarization ensures that each subnet is advertised with its correct subnet mask, allowing precise routing between non-contiguous subnets.

4. Verify RIP routing with the following commands:

- `show ip protocols` – to confirm RIP v2 is active.
- `show ip route` – to verify RIP routes are installed.
- `debug ip rip` – to monitor RIP updates in real-time (optional).